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Author(s): Rasyad A. Parinduri, Yoong Hon Lee and Kung Ming Tiong
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The effects of bigger rewards in individual tournaments on efforts and risk taking: evidence from chess

By Rasyad A. Parinduri,[a] Yoong Hon Lee,[a] and Kung Ming Tiong[b]

[a]Nottingham University Business School, University of Nottingham Malaysia Campus, Jalan Broga, Semenyih, Selangor, 43500, Malaysia; e-mail: lee.yoong-hon@nottingham.edu.my
[b]Centre for English Language and Foundation Education, University of Nottingham Malaysia Campus

Abstract

We examine the effects of bigger rewards in individual dynamic tournaments on efforts and risk taking using the three-point rule in chess. Most chess tournaments use the standard rule while some tournaments use the Bilbao rule, which is identical to the three-point rule in soccer: We observe the same pairs of chess players playing under both rules, a research design that fits fixed-effect models. We find the Bilbao rule makes games 33% more decisive, mostly to white players' advantage, who win 50% more games. We identify two mechanisms why the Bilbao rule works: It encourages players to play longer and discourages them from using drawish openings. These results suggest incentive schemes that provide bigger rewards for better performances work in individual multi-stage tournaments in which efforts and financial rewards are directly linked, and in which strategic interactions among teammates and with competitors are less complex.

JEL classifications: J22, J33, J41, M52.

1. Introduction

Some markets feature tournament-like competition in which rewards depend on relative, rather than absolute, performance. Workers compete for promotion; sales agents secure more deals to win bonuses; companies compete in patent races; students study hard to graduate with distinction; politicians beat each other to win majority votes in elections.

While many papers have examined how people behave in tournaments (Ehrenberg and Bognanno, 1990; Knoeber and Thurman, 1994; Chevalier and Ellison, 1997) and in other performance evaluation schemes (Oyer, 1998; Lazear, 2000), few have examined whether tournament-rule changes that increase rewards for good performances affect efforts and

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risk taking—even fewer look at the effects on individual performances.[1] Papers that examine the effects of rule changes usually use sports data such as Guedes and Machado (2002), Aylott and Aylott (2007), Dilger and Geyer (2009), and Moschini (2010). Using soccer data, they find mixed evidence on whether bigger rewards for good team performances matter. The effects of bigger rewards for good individual performances may differ, however.

Moreover, few papers examine the effects of tournament-rule changes in dynamic, multi-stage, tournaments. Employees participate in annual performance reviews whose results may determine promotion; students compete to get A grades, which determines their degree classifications. The empirical literature is scant on, for example, what happens if companies decide to prioritize annual top performers for promotion, or what happens if universities assign more weight in GPA calculations to marks obtained by students in advanced modules.

This paper, using fixed-effect models, examines the effects of a rule change in chess—the three-point rule—that increases rewards for good individual performances in a dynamic tournament. Most chess organizers have always used the standard rule, which rewards one point for a win, 0.5 for a draw, and zero for a loss; but recently several organizers have used a new rule called the Bilbao rule, which rewards three points for a win, one for a draw, and zero for a loss. (The Bilbao rule therefore resembles the three-point rule in soccer: It increases the value of a win from the value of two draws to that of three draws.) We control for factors that may affect players' behaviour and their participation in tournaments that use the Bilbao rule, which include player, round, and year fixed-effects. Hence, we compare similar games (i.e. games played by the same pairs of players, in the same rounds relative to the last round, whose ratings and numbers of wins (at the start of the round) were similar, and so on) that differ only in one respect: Some games are played under the standard rule while others are played under the Bilbao rule. If we see the outcomes of games under the Bilbao rule differ from those under the standard rule, we can attribute the difference to the Bilbao rule's extra point for a win.

We contribute to the growing empirical literature on tournaments in three ways.[2] Firstly, we offer some evidence on the effects of bigger rewards for good individual performances in a dynamic tournament, which may shed light on its effects in individual tournaments in other markets. Secondly, analysing chess allows us to avoid team sports' coordination and free-riding problems that may confound the effects of bigger rewards for good performances. Thirdly, we can control for many player and game characteristics, which makes games under Bilbao and standard rules more comparable and provides us with unbiased estimates of the effects.

Intuitively, awarding more points for a win encourages aggressive plays and makes games more decisive but empirical evidence on the effects of the three-point rule in soccer is mixed, which raises the question of whether one extra point for a win is powerful enough

1 This line of literature follows Lazear and Rosen (1981), Green and Stokey (1983), and Nalebuff and Stiglitz (1983).

2 Recent empirical papers on tournaments are, among others, Carpenter *et al.* (2010) on tournaments and office politics; Freeman and Gelber (2010) on prize structure and information; Dohmen and Falk (2011) on sorting; Genakos and Pagliero (2012) on interim rank; Güth *et al.* (2016) on output-dependent prize; Sato (2016) on risk taking among mutual fund managers; and Shen and Zhang (2018) on incentives for promotion and firm innovation.

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an incentive to change players' behaviour.[3] Aylott and Aylott (2007), Dilger and Geyer (2009), and Moschini (2010), for example, find the three-point rule increases the number of goals and decreases the proportion of drawn games; Guedes and Machado (2002), however, find the rule works for only underdog teams; Palacios-Huerta (2004) show that it does not seem to affect the average number of goals. Lee and Parinduri (2016), using regression discontinuity design that resembles randomized experiments, find the effects of the three-point rule are subtle: It does not make games more decisive or increase the number of goals but it increases the second-half goals of teams losing at half-time.

Perhaps the three-point rule matters more, provides stronger incentives, in individual games like chess in which players' efforts are easily observed and directly linked to financial rewards.[4] In team sports, game dynamics and strategic interactions among teammates and with opponent players are clearly more complex. Some soccer players in a team, for example, may not respond to the three-point rule as strongly as their coaches want them to (which makes the rule less effective) or they may free-ride on the efforts of their teammates—problems that do not exist in chess. A leading soccer team may turn defensive to maintain the lead and still win; a chess player whose position is better cannot just maintain the lead—if she wants to win the game, she has to keep fighting until her opponent resigns.[5] Soccer players' salaries do not exclusively depend on whether their team wins a game or becomes the champion; financial rewards are not equally distributed among players either (players' contracts may differ from one another). A chess player gets the entire prize money for herself if she wins a tournament.

We use Guedes and Machado (2002) to guide our analyses of the effects of bigger rewards for good performance on risk taking. They argue that—if we apply their theory in the context of chess—the Bilbao rule may induce weaker players to play more defensively when they compete against very strong players. Because white players have the first-mover advantage, their theory also implies that, under the Bilbao rule, white players play more aggressively (and choose riskier openings) while black players play more defensively.

We find the Bilbao rule induces players to fight harder: It makes games 33% more decisive, mostly to white players' advantage, who win 50% more games. We do not find evidence that the Bilbao rule increases black players' likelihood of winning or losing games, probably because, in games played by top grandmasters, black players initially try to neutralize white players' first mover advantage and only fight to win when white players make inaccurate moves. However, as white players play more aggressively, the Bilbao rule is likely to press black players to work harder to secure a draw. Overall, therefore, the Bilbao rule induces players to play fighting chess and makes games more decisive.

We identify two mechanisms why the Bilbao rule works. It makes players play longer (by four to five moves, about 10% longer): Players work harder, take more risks and, when

3 This line of literature relates to that on the use of tournaments and piece-rate incentive contracts in organizations to elicit greater employee efforts (Lazear and Rosen, 1981; Lazear, 2000; Shearer, 2004).

4 As far as we know, no theoretical paper analyses whether the three-point rule matters in individual games like chess, but theoretical papers on soccer show its effects are also subtle; see, for example, Guedes and Machado (2002), Brocas and Carrillo (2004), Haugen (2008), and Moschini (2010).

5 Chan *et al.* (2009) show, in team sports, once a team enjoys a comfortable margin of victory, the team's efforts may drop.

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opportunities arise, try to squeeze a win. The rule also discourages players from using drawish openings (by about 23% if we define drawish openings as ones with 40% draws in past games), which makes games more likely to be imbalanced and double-edged.[6] (We define an opening as a drawish opening if its percentage of drawn games in the past is high.)

2. Draws in chess and the Bilbao rule

Draws are a logical outcome of chess games: If a white and a black player in a chess game accurately play, the game is likely to end in a draw.[7] That is why draws are common in chess, especially among top players with similar ratings. In the 2016 Fédération Internationale des Échecs (FIDE) World Chess Candidates Tournament, which determined the challenger to the world champion Magnus Carlsen, 71% of the games were draws. Historically two in five games among players in the top 100, and one in two games among players in the top 10 whose rating difference is at most 400 points, were draws (Sonas, 2011).

Fans enjoy watching hard-fought drawn games, but they loathe short unfought draws agreed by players called 'grandmaster draws'. From chess fans' perspective, a grandmaster draw is like a boxing match in which the boxers keep hugging each other, declare a truce before the first round ends, and leave the ring.[8] The problem is that grandmaster draws are common: One in 10 chess games among players in the top 10 were drawn prior to move 25 (Sonas, 2011). A notorious example is the last game of the Man vs Machine Match between Kasparov and Deep Junior in 2003. When Kasparov had chances to win the game, and the match, he offered a draw on move 23 to the disappointment of millions of fans watching the game (Ashley, 2003). (The Deep Junior team rejected the offer but the team returned it five moves later, and Kasparov accepted.) In the Kasparov–Anand Professional Chess Association (PCA) World Championship final in 1995, the first eight games were draws and only one of them lasted longer than 30 moves; when Kasparov led by three wins after 14 games (with six games remaining), the following two games lasted only 20 and 16 moves, respectively; in round 18, Anand agreed to a draw after only 12 moves.

Players may agree to have grandmaster draws for strategic reasons. Leading players in a tournament may try to preserve their positions because the positions are tied to financial rewards. High-rated players may want to avoid the risk of losing games against low-rated players because they would lose many rating points; low-rated players may agree to short draws against high-rated players just to gain rating points or to secure grandmaster norms.[9] Players may feel exhausted in the middle of a tournament and agree to short draws to give

6 Three of the most drawish openings are Lasker's Queen's Gambit Declined, Tartakower's Queen's Gambit Declined, and Queen's Indian; three of the sharpest openings are Kings Knight Gambit, Sicilian Modern Dragon, and Two Knights.

7 A draw occurs when a player has no legal move but is not in check (stalemate); players repeat the same position three times and one of them claims a draw (threefold repetition rule); players do not capture any piece or move any pawn in the last 50 moves and one of them claims a draw (the 50-move rule); no player has enough materials to checkmate the other; or players agree to draw the game (a draw by agreement).

8 Some chess players, however, think draws should be allowed at any stage of the game; see, for example Association of Chess Professionals (2016).

9 A norm is an indicator of an excellent performance in a chess tournament. To be a grandmaster, a player has to secure several norms.

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them more time to recuperate, or they may have nothing to play for in later rounds because tournament leaders are impossible to catch up with.

Members of the chess community have proposed solutions to the problem of short draws but one of the few that have been tried so far is the Bilbao rule, which increases rewards for a win.[10] It was first introduced as an experiment in the Bilbao Blindfold Chess World Cup in 2007 (ChessBase, 2007) to induce players to play attacking and entertaining chess, and to discourage them from drawing games in the last rounds of tournaments (to guarantee their standings in the tournaments and, hence, prize money). Under the standard rule, a win, a draw, and a loss are worth 1, 0.5, and 0 points, respectively; under the Bilbao rule they are worth 3, 1, and 0 points—the Bilbao rule therefore resembles FIFA's move from the two-point rule to the three-point rule. Under the standard rule, players trailed by just one point (i.e. a win) in the last few rounds of a tournament may find it difficult to catch up with tournament leaders (because the leaders could easily secure draws by agreeing to short draws or playing drawish openings); under the Bilbao rule, trailing players find it easier to overtake leaders if they chalk up enough wins.

3. Empirical strategy and data

3.1 Empirical strategy

We estimate the effects of the Bilbao rule on how players behave using the following fixed-effect model:

$$y_{ijrty} = \alpha + \beta D_{ty} + \sum \gamma_{rty} Round_{rty} + \sum \delta_{i/j\,rty} Player_{i/j\,rty} + \xi_y + \varepsilon_{ijrty} \quad (1)$$

where y_{ijrty} is a measure of outcomes of a game between white player i and black player j in round r of tournament t in year y; D_{ty} is an indicator equals one if the game was played under the Bilbao rule; *Round* is the round characteristics of the game, which include a set of dummy variables for each round relative to the last round of each tournament (i.e. ξ_r, dummy variables for the last round, the penultimate round, the third-last round, etc.) to allow games played in later rounds to have higher importance; *Player* is the characteristics of the players, which include a quadratic function of player i's rating and that of player j's, the number of wins each player had at the start of the round, the difference between the number of wins each player had and the number of wins of the highest-ranked player at the start of the round, and player i and j fixed-effects (ξ_i and ξ_j); ξ_y is year fixed-effects; and ε is the error terms. (We cannot include tournament fixed-effects as control variables because the Bilbao rule had been used by the same tournaments during the period of analysis.)

Identification relies on the assumption that, after controlling for the round characteristics, player characteristics, and year fixed-effects, the games are comparable except that some games were played under the Bilbao rule while some others were not. We therefore compare games played by the same pairs of players, whose ratings and number of wins (at the start of the round) were similar, in the same rounds relative to the last round, and so on. To the extent that we have controlled for important factors that might affect players' behaviour and players' participation in tournaments that used the Bilbao rule, the

10 Others propose to reward wins with money, forbid draws by agreement (Sofia rule), or forbid draw offers before a certain number of moves; others call on tournament organizers to invite only attacking players. See, for example, Nunn (2005), ChessBase (2007), and Shipov (2011).

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coefficient of D_{ty} in eq. (1), β, would tell us about the effects of the Bilbao rule on players' behaviour and game outcomes.

We explore whether the effects of the Bilbao rule in more important games—games in later rounds played by players who were still in contention to win tournaments—differ. We estimate:

$$y = \alpha + \beta D + \theta Leaders + \vartheta LastRounds + \mu D * Leaders + \pi D * LastRounds + \rho D * Leaders * LastRounds + \sum \gamma Round + \sum \delta Player + \xi + \varepsilon \quad (2)$$

where *Leaders* is an indicator equals one if a player had 0.5-win (or 1-win) difference from the leaders' and zero otherwise; and *LastRounds* is an indicator equals one if a round was the last two (or three) rounds. The coefficients of the interaction terms would tell us whether the Bilbao rule matters more in crucial games between leaders than it does in less important games among laggards.

We also examine whether players respond differently to the Bilbao rule if they play against stronger or weaker players. We estimate:

$$y = \alpha + \beta D + \theta RatingDifference + \mu D * RatingDifference + \sum \gamma Round + \sum \delta Player + \xi + \varepsilon \quad (3)$$

where *RatingDifference* is the difference between white and black players' ratings.

3.2 Data

We get the data from *The Week in Chess* (TWIC), a weekly chess magazine that reports detailed information on games played in major chess tournaments around the world.[11] The data include information on players' ratings, tournament types and categories, outcomes of each game, number of moves in each game, and openings played by players. To ensure the data are accurate, we cross-check the TWIC data against game archives of chessgames.com, an online database of chess games.

To make the games comparable, we include games played in elite tournaments from 2006 to 2014 in which only top grandmasters played. (We exclude tournaments in earlier years because the Bilbao rule was used for the first time in the Bilbao tournament in 2008.) We have five tournaments in the sample: Bilbao Masters, London Chess Classic, Tal Memorial, Dortmund, and Corus/Tata Steel. The Bilbao Masters and London Chess Classic used the Bilbao rule; the others used the standard rule. The tournaments were either single- or double-round robin; almost all used the classical time control.[12] We include only editions of tournaments that used the classical time control (we exclude the London Chess Classic in 2013 and Tal Memorial in 2014 because they used rapid or blitz time controls).[13]

11 The website is http://theweekinchess.com. We download the data on 17 June 2015.

12 We exclude a few other elite tournaments (e.g. Linares, Mtel Masters, Zurich Chess Challenge, Norway Chess, Grenke Chess Classic, Sinquefield Cup, Gashimov Memorial, Biel Chess Festival, and Capablanca Memorial) because they were knockout type tournaments, discontinued early on, recently introduced, or Category 18 or lower.

13 In classical time control, each player has 90 minutes to make the first 40 moves and 30 minutes extra for the rest of the game and 30 seconds per move starting from move one. Rapid time controls refer to short time control (for example, 25 minutes with a 10-second increment after each move). Blitz time controls refer to very short time control (for example, three minutes with a two-second increment after each move).

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Table 1. Tournaments we include in the sample

Tournament	Time period	Scoring format	Number of games
Bilbao Masters	2008–2014	Bilbao	138
London Chess Classic	2009–2014	Bilbao	139
Tal Memorial	2006–2014	Standard	360
Dortmund	2006–2014	Standard	292
Corus/Tata Steel	2006–2014	Standard	794

Notes: The number of all games is 1,723; the number of games under the Bilbao rule is 277.
Source: Authors' calculations.

The average players' ratings in these five tournaments were all above 2,700 (Category 19 and above) except for Dortmund 2008 and London Chess Classic 2009 whose average ratings fell a few points below 2,700. There are 1,723 games in the sample; about one in six were played under the Bilbao rule (see Table 1).

In the basic specifications, we use the outcomes of games to measure players' willingness to play attacking chess: *decisive games*, *white wins*, and *white's points*. The first two are dummy variables equal one if a game is decisive and white player wins, respectively; *white's points* is the point awarded to white players under the standard rule. If players play more aggressively, white and black players are more likely to play double-edged games, which leads to decisive games. In particular, if white players purposely play to win, there may be more white's wins, and white players will get more points. (Because black players have no first-mover advantage, they usually play to equalize and only later to win if white players play inaccurate moves.)

We also use the length of games to measure how hard players try to press for a win. We use three measures of outcomes: *number of moves*, *30 moves or more*, and *40 moves or more*. The first is the number of moves by white players; the second and the third are dummy variables equal one if a game has 30 and 40 moves or more, respectively. We use *30 moves or more* as a measure of outcome because it indicates games are played long enough; we use *40 moves or more* as a measure of outcome because it indicates players hit the first time control (at move 40).

We also use the types of openings played by players as a measure of outcome—whether they are drawish or double-edged. Opening moves in chess are pivotal to game outcomes. Some openings are very safe for white: They lead to quiet and uncomplicated games, but they are drawish; other openings are more aggressive, which may lead to double-edged games in which white players keep a slight advantage and black players fight to equalize, though the latter may take over the initiative if white players make errors. Whether a white player chooses a particular opening depends on how he thinks black would respond to that opening, which the player analyses before the game is played. (Every chess player has access to all games his opponent played in the past, which gives him some idea on whether the opponent is an expert of an opening and the opponent's likely choice of variation in the event that he plays that particular opening.) Therefore, if a white player wants to push for a win, he needs to avoid drawish openings; he needs to use an aggressive opening that his opponent has no expertise in.

We use the percentage of drawn games and that of short drawn games in the past as a measure of whether an opening is drawish. To construct these variables, for each of the

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openings used by players in games in our data, we find out its percentage of drawn games in the past on chessgames.com.[14] We consider two sets of games: all games in the database and games played since 1980 (to allow the possibility that chess theories might have changed in the past few decades). We create six measures of drawish openings: openings whose percentage of draws are higher than 40% in all games and those higher than 45%; We create six measures of drawish openings: openings whose percentage of draws are higher than 40% in games played since 1980 and those higher than 45%; openings whose percentage of short draws are higher than 15% in all games and those higher than 20%. (The openings have 40% draws and 15% short draws on average.) All are dummy variables; for example, *percentage of drawn games more than 40%* is a dummy variable equals one if an opening leads to more than 40% draws in the past.

If the Bilbao rule induces players to play more aggressively, we expect to see the following in the data: (i) games are more decisive and white players win more games, (ii) games are longer, and (iii) players are less likely to use drawish openings.

The summary statistics in Table 2 indicate these effects. Under the Bilbao rule, games were six percentage points more decisive and white (black) players were four (three) percentage points more likely to win; on average white's points did not differ, however (Panel A). Games played under the Bilbao rule were five moves longer; they were about 15 percentage points more likely to go beyond 30 or 40 moves (Panel B). Players who played under the Bilbao rule were less likely to play drawish openings: five percentage points less likely to play openings whose percentage of drawn games are more than 40% or 45% (Panel C). Players were two percentage points more likely to use openings whose percentage of drawn games are more than 50%, however.

The ratings of players in tournaments that used the standard and Bilbao rule were similar, but the two types of tournaments had some differences. Tournaments that used Bilbao rule on average had one fewer round; they also had about five fewer players on average (Panel D). Tournaments were more competitive under the Bilbao rule in the sense that more players had at most 0.5 or 1-win difference from the leader's (Panel E).

4. Results

Firstly, we examine whether the Bilbao rule makes games more decisive. Secondly, we analyse two mechanisms why games may become more decisive: whether the rule lengthens the games and whether it induces players to avoid drawish openings. Then we explore whether the rule matters more in more important games (games played in the last few rounds by players who have realistic chances to win tournaments).

4.1 Do games become more decisive?

Table 3 shows that the Bilbao rule seems to make games more likely to be decisive (or less likely to end in a draw): All estimates are both large in magnitude and statistically significant. Compared to games under the standard rule, games under the Bilbao rule are six percentage points more likely to be decisive (column 1). When we include round characteristics (a set of dummy variables for each round relative to the last round of each

14 In the data, the players used 278 openings. For consistency, we convert two openings in the list: We change 'A05 Various' to 'Reti Opening' and 'A13 Reti' to 'English opening'. We downloaded the data on 17 June 2015.

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Table 2. Summary statistics

	Standard rule (1)	Bilbao rule (2)
A. Game outcomes		
Decisive games	0.391	0.455
	(0.488)	(0.499)
White players win	0.253	0.289
	(0.435)	(0.454)
Black players win	0.138	0.166
	(0.345)	(0.373)
White's points	0.558	0.561
	(0.307)	(0.332)
B. Length of games		
Number of moves	43.151	48.780
	(17.790)	(19.205)
30 moves or more	0.777	0.935
	(0.416)	(0.247)
40 moves or more	0.528	0.668
	(0.499)	(0.472)
C. Drawish openings		
Percentage of drawn games more than 40%	0.482	0.433
	(0.500)	(0.496)
Percentage of drawn games more than 45%	0.167	0.202
	(0.373)	(0.402)
D. Players' ratings, number of rounds, and number of players		
White's ratings	2735.048	2753.960
	(50.067)	(57.266)
Black's ratings	2735.454	2754.801
	(49.078)	(56.676)
Number of rounds	6.006	4.534
	(3.399)	(2.510)
Number of players	11.701	6.794
	(2.580)	(1.758)
E. Groups of games		
Games are played in last two rounds	0.189	0.260
	(0.392)	(0.439)
Games are played in the last three rounds	0.284	0.390
	(0.451)	(0.489)
A player has at most 0.5-win difference from the leader's	0.641	0.791
	(0.480)	(0.408)
A player has at most 1-win difference from the leader's	0.833	0.917
	(0.373)	(0.276)

Notes: The number in each cell is the proportion or mean. All variables except *white's points*, *number of moves*, *ratings*, *number of rounds*, and *number of players* are dummy variables. The figures in parentheses are standard deviations.
Source: Authors' calculations.

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Table 3. The Bilbao rule makes games more decisive

Dependent variable: *Decisive games*	(1)	(2)	(3)	(4)	(5)
Bilbao rule	0.064*	0.112**	0.115**	0.117*	0.131**
	(0.033)	(0.037)	(0.038)	(0.049)	(0.050)
Control variables					
Round characteristics		✓	✓	✓	✓
Quadratic function of white and black players' ratings			✓	✓	✓
White and black players fixed-effects				✓	✓
Year fixed-effects					✓
Number of observations	1,723	1,708	1,708	1,708	1,708
Adjusted R^2	0.002	0.012	0.034	0.081	0.083

Notes: The number in each cell is the estimate of the effects of the Bilbao rule on whether games are decisive with or without sets of control variables listed in the bottom rows. *Bilbao rule* equals one if a game is played under the Bilbao rule and zero otherwise; *decisive games* equals one if a game is decisive. Round characteristics include a set of dummy variables for rounds (the last round, second-last round, etc.), players' number of wins before the start of the round, and the difference between the number of wins each player had and the highest ranked player's before the round started. The figures in parentheses are robust standard errors. One and two stars indicate statistical significance at a level of 5% and 1%, respectively.
Source: Authors' calculations.

tournament) as control variables (column 2), we find the Bilbao rule increases the likelihood that games are decisive by 11 percentage points, an estimate that we also get when we add player characteristics (a quadratic function of white's rating and that of black's, the number of wins each player had before the start of the round, and the difference between the number of wins each player had and the number of wins of the highest ranked player before the round started—each of the last two is a set of dummy variables) and some fixed-effects (white's and black's fixed-effects and year fixed-effects) as controls (columns 3–5). Using the most complete specification (column 5), we find the Bilbao rule increases the likelihood that games are more decisive by 13 percentage points—a 33% increase.[15]

The Bilbao rule makes games more decisive but to whose advantage? White players, Table 4 suggests: White have more wins and get more points (in these specifications, we calculate the points under the standard rule). The Bilbao rule increases the probability of white wins by 8 and 13 percentage points when we control for only round and both round and player characteristics, respectively (columns 1–2). It does not seem to make black players more likely to win or lose, however: The estimates (columns 3–4) are statistically insignificant; it is virtually zero when we control for both round and player characteristics (column 4). Because white players win more games and have fewer draws, white players get slightly more points on average (column 6).

15 We cannot control for all tournament rules such as whether the tournaments use the Sofia rule, the particulars of the time control, tie breaks, playoff rules, and monetary prizes because they are not available for each of the tournaments in each of the years they were organized; however, we think our results are quite robust.

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Table 4. The Bilbao rule increases the likelihood that white players win

Dependent variable	White win		Black win		White's points	
	(1)	(2)	(3)	(4)	(5)	(6)
Bilbao rule	0.081*	0.129**	0.032	0.002	0.025	0.063*
	(0.032)	(0.041)	(0.025)	(0.034)	(0.022)	(0.028)
Control variables						
Round characteristics	✓	✓	✓	✓	✓	✓
Player characteristics		✓		✓		✓
Adjusted R^2	0.138	0.238	0.125	0.155	0.208	0.282

Notes: The number in each cell is the estimate of the effects of the Bilbao rule on whether white players win, black players win, and points obtained by white under the standard rule with and without a set of control variables. *Bilbao rule* equals one if a game is played under the Bilbao rule and zero otherwise; *white win* and *black win* are dummy variables; *white's points* equals one if white players win, zero a loss, and 0.5 a draw. See also the notes to Table 3. *, ** indicates statistical significance at a level of five and one percent, respectively.
Source: Authors' calculations.

The results in Tables 3 and 4 indicate the Bilbao rule induces white players to play more aggressively and exploit their first mover advantage, pressurizes black players to work harder to secure a draw, and makes games more exciting overall. Under the Bilbao rule, white players have more wins (columns 1–2 of Table 4), which indicates the Bilbao rule induces white players to play harder to win. It does not seem to affect black players' likelihood to win or lose, however (columns 3–4 of Table 4), which means, given white players' first-mover advantage and more aggressive plays, black players have to work harder to secure a draw. The Bilbao rule, therefore, induces both white and black players to play fighting chess (Table 4) and makes games more likely to be decisive (Table 3).

4.2 Does the Bilbao rule lengthen games and discourage drawish openings?

One mechanism why games become more decisive is players play longer to press for a win, which Table 5 seems to suggest. On average, the Bilbao rule increases the number of moves by four or five (i.e. eight to 10 plies) depending on whether we include player characteristics as controls (columns 1–2), which is about 10% increase. Games are therefore more likely to go beyond 30 moves (by 11 percentage points, or 14%, if we use the most complete specification (column 4)); they are also more likely to go 40 moves or more though the estimate is statistically significant when we include only round characteristics as control (columns 5–6). Players seem to prefer to battle it out rather than to settle for a quick draw when they play in tournaments that use the Bilbao rule.

As Table 6 shows, another mechanism at play is that players are more likely to avoid drawish openings. The Bilbao rule decreases the likelihood that players play drawish openings by about 11 percentage points, or 23%, when we define drawish openings as openings that have more than 40% draws in the past (row A, column 1); there is no evidence that they are less likely to use openings whose drawn games are more than 45% (row B, column 1). None of the estimates is statistically significant, however, if we use games played since 1980 to define drawish openings both in terms of percentage of draws and percentage of short draws (rows C–F, column 1). If we drop white and black players' fixed-effects from the regressions (to increase the statistical power to reject the null hypotheses), the estimates

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Table 5. The Bilbao rule makes games longer

Dependent variable	Number of moves		30 moves or more		40 moves or more	
	(1)	(2)	(3)	(4)	(5)	(6)
Bilbao rule	5.019**	3.554*	0.142**	0.107**	0.127**	0.046
	(1.357)	(1.777)	(0.024)	(0.035)	(0.037)	(0.050)
Control variables						
Round characteristics	✓	✓	✓	✓	✓	✓
Player characteristics		✓		✓		✓
Adjusted R^2	0.017	0.067	0.023	0.083	0.018	0.073

Notes: The number in each cell is the estimate of the effects of the Bilbao rule on the number of moves or the probability that a game goes beyond 30 or 40 moves with or without a set of control variables. *Bilbao rule* equals one if a game is played under the Bilbao rule and zero otherwise; *30 moves or more* and *40 moves or more* are dummy variables. See also the notes to Table 3. *, ** indicates statistical significance at a level of five and one percent, respectively.
Source: Authors' calculations.

are statistically significant if we define drawish openings using 40% draws or 15% of short draws (column 2).

4.3 Does the Bilbao rule matter more in crucial games?

We explore whether the Bilbao rule matters more in games played in later rounds by players who have chances to win tournaments (eq. [2]), but we do not find evidence of this. (We do not present the results for the sake of brevity; they are available upon request.) We consider two ways to define the later rounds—the last two rounds and the last three rounds—and two ways to define contesting players—0.5 win difference and 1-win difference from the tournament leader's number of wins. Regardless of which definition we use, we find statistically insignificant estimates. These results seem to suggest that the Bilbao rule does not differently affect games in later rounds (it affects games in all rounds similarly), but given that the estimates are imprecise (the standard errors are large, as large as the estimates if not larger) it is also possible that we lack sufficient statistical power to differentiate the effects of the Bilbao rule on games in earlier and later rounds.

4.4 Does rating difference induce players to respond differently?

We examine whether players respond differently to the Bilbao rule on key outcome variables if their opponents are stronger or weaker (eq. [3]) but we do not find evidence that they do (Table 7). The estimates of the direct effect of the Bilbao rule have similar magnitude and are statistically significant but none of the interaction terms are (column 1). If we drop white and black players' fixed-effects from the regressions, the interaction terms for regressions on the number of moves and the likelihood of using drawish openings are statistically significant (column 2). The first is negative and the second is positive: The Bilbao rule decreases the number of moves by five moves if white player's rating is 100 higher; the same rating difference induces the players to be more likely to use drawish openings by a 0.1 percentage point, a very small increase.

4.5 Does Sofia rule confound the results?

We should have controlled for whether a tournament used the Sofia rule but unfortunately we do not observe this information for each tournament in each of the years they were organized.

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Table 6. The Bilbao rule discourages the use of drawish opening

Dependent variable: Drawish openings	(1)	(2)
A. >40% draws (in all games)		
Bilbao rule	−0.105*	−0.141**
	(0.051)	(0.039)
B. >45% draws (in all games)		
Bilbao rule	0.030	0.001
	(0.042)	(0.031)
C. >40% draws (in games played since 1980)		
Bilbao rule	−0.077	−0.114**
	(0.051)	(0.040)
D. >45% draws since 1980 (in games played since 1980)		
Bilbao rule	−0.034	−0.019
	(0.046)	(0.035)
E. >15% short draws (in games played since 1980)		
Bilbao rule	−0.093	−0.119**
	(0.049)	(0.039)
F. >20% short draws (in games played since 1980)		
Bilbao rule	0.003	0.004
	(0.043)	(0.032)
Control variables		
Round and player characteristics	✓	✓
White and black players fixed-effects	✓	

Notes: The number in each cell is the estimate of the effects of the Bilbao rule on a dependent variable listed in the left column. See also the notes to Table 3. *, ** indicates statistical significance at a level of five and one percent, respectively.
Source: Authors' calculations.

We address this lack of control variable by running regressions on key outcome variables using games whose number of moves are more than 30 and, to a large extent, we find similar results. We find the estimates have similar sign and magnitude as those in Tables 3–6 but they are statistically insignificant at 5% level (most are statistically significant at 10% level). We think that the estimates are statistically insignificant at 5% level because: (i) these estimates rely on a presumably small variation in the Bilbao rule across the same pair of white and black players, and (ii) we have a smaller sample size—both lead to lower statistical power to reject the null hypotheses. We therefore estimate the same regressions without white and black players fixed-effects: We find the signs and magnitude of the estimates are similar; the effects on whether the Bilbao rule makes games more decisive is statistically significant at 5% level while that of whether players use drawish openings is statistically significant at 1% level. (We do not present the results for the sake of brevity.)

4.6 Does the Bilbao rule reduce competitive balance?

We check whether the results come at the cost of competitive balance but we find tournaments that use the Bilbao rule are more competitively balanced. We calculate the competitive balance of a tournament as the standard deviation of points accumulated by players at the end of the tournament divided by $(0.5/\sqrt{n})$ where n is the number of games played by

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Table 7. Does rating difference induce players to respond differently?

Dependent variable	(1)	(2)
A. Decisive games		
Bilbao rule	0.131**	0.120**
	(0.050)	(0.039)
Bilbao rule * rating difference	0.0001	−0.0001
	(0.001)	(0.0004)
B. White win		
Bilbao rule	0.128**	0.080*
	(0.041)	(0.032)
Bilbao rule * rating difference	0.001	0.0002
	(0.001)	(0.0003)
C. Number of moves		
Bilbao rule	3.606*	4.093**
	(1.789)	(1.388)
Bilbao rule * rating difference	−0.043	−0.048*
	(0.030)	(0.019)
D. Drawish openings (> 40% draws)		
Bilbao rule	−0.107*	−0.142**
	(0.051)	(0.039)
Bilbao rule * rating difference	0.001	0.001*
	(0.001)	(0.0004)
Control variables		
Round and player characteristics	✓	✓
White and black players fixed-effects	✓	

Notes: The number in each cell is the estimate of the effects of the Bilbao rule and its interaction with white and black players' rating difference on a dependent variable listed in the left column. See also the notes to Table 3. *, ** indicates statistical significance at a level of five and one percent, respectively.
Source: Authors' calculations.

each player (Fort and Quirk, 1995). Figure 1, which presents the distribution of competitive balance in tournaments by tournament type, shows that tournaments that use the Bilbao rule are more skewed to the right. The averages in Bilbao and non-Bilbao tournaments are 3.6 and 5.5, respectively.

5. Conclusion

The Bilbao rule induces players to play fighting chess and makes chess games more exciting, which would delight fans and tournament organizers. 'I've played [tournaments that use the Bilbao rule] before, and it's nice to know that you can always … come back', said Magnus Carlsen after losing a game in an earlier round, won two games in a row, and re-took the lead in Bilbao Masters 2016. The rule, because it rewards more points for a win, reduces the likelihood of drawn games and makes games 33% more decisive—results that we argue are causal estimates of the effects because we observe the same pairs of players playing under the standard rule and under the Bilbao rule, which allows us to control for many factors that may confound identification. We do not find evidence the rule matters

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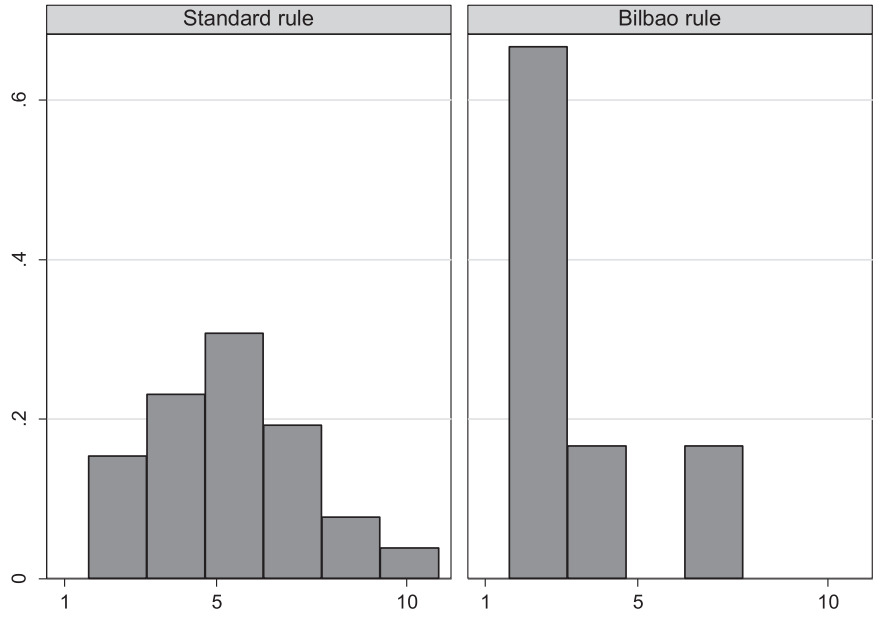


Fig. 1. Distribution of the tournaments' competitive balance.
Notes: We calculate the competitive balance of a tournament as the standard deviation of points accumulated by players at the end of the tournament divided by $(0.5/\sqrt{n})$ where n is the number of games played by each player in the tournament (Fort and Quirk, 1995).
Source: Authors' calculations.

more in games played in later rounds by players who have some chance to be tournament champions (which means either the rule matters equally in all rounds or we lack sufficient statistical power to reject the null hypothesis).

We identify two possible mechanisms: Players play longer, and they are less likely to choose drawish openings. By playing longer, players think harder and take more risks but, in return, they give themselves more opportunities to eke out wins when their positions allow them to. By avoiding drawish openings, games are more likely to become double-edged, which makes players more likely to make inaccurate moves and leads to imbalanced positions and, later, decisive outcomes.

These findings show that incentive schemes like the three-point rule do work in some sports and, perhaps, in individual tournaments in other markets: In contrast to mixed evidence in the literature on its effects in soccer (Guedes and Machado, 2002; Palacios-Huerta, 2004; Aylott and Aylott, 2007; Dilger and Geyer, 2009; Moschini, 2010; Lee and Parinduri, 2016), the three-point rule works in chess.[16] Perhaps the rule provides stronger incentives in individual games in which efforts and financial rewards are directly linked and game dynamics and strategic interactions among teammates and with opponents are less complex.

The literature suggests that participants in tournaments exert more effort and make riskier moves when the stakes are higher (Ehrenberg and Bognanno, 1990; Becker and Huselid, 1992; Coupé *et al.*, 2006; Freeman and Gelber, 2010); our results show that this holds in

16 It is also possible, however, that the three-point rule in chess matters because chess players have higher overall cognitive ability than athletes in general (Sala *et al.*, 2017).

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dynamic, multi-stage, tournaments too. For example, in the context of annual performance reviews and promotion of employees, our results suggest that employees may work harder, and take annual performance reviews more seriously, if human resource departments prioritize annual top performers for promotion; students may study harder in more advanced modules if universities assign more weight to these modules in GPA calculations; politicians may increase campaign expenses in later rounds of elections that require a simple majority to win. The results on risk taking mean employees may be more willing to take difficult jobs, or adopt riskier strategies, to become annual top performers and get promoted; students may be more willing to take difficult or large-credit modules as electives to graduate with distinction.

Because the effects of bigger rewards for good performances differ in individual and team sports, organizers of team sports such as soccer, or team tournaments in other markets such as patent races, could continue tinkering with their scoring systems to improve team performances. Leagues of team sports, in particular, have hopes to make their sports more exciting, without changing game rules too much, by adding sufficiently strong incentives for a win.

Supplementary material

Supplementary material is available on the OUP website. This comprises the data and replication files, and an Online Appendix.

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References

Ashley, M. (2003) The end of the draw offer? http://www.thechessdrum.net/65thSquare/DrawOffer/DrawOffer3.html (accessed 14 August 2018).

Association of Chess Professionals. (2016) ACP Survey 2016, http://www.chessprofessionals.org/sites/default/files/ACP Survey 2015.pdf (accessed 14 August 2018).

Aylott, M. and Aylott, N. (2007) A meeting of social science and football: measuring the effects of three points for a win, *Sport in Society*, **10**, 205–22.

Becker, B.E. and Huselid, M.A. (1992) The incentive effects of tournament compensation systems, *Administrative Science Quarterly*, **37**, 336–50.

Brocas, I. and Carrillo, J.D. (2004) Do the 'three-point victory' and 'golden goal' rules make soccer more exciting? *Journal of Sports Economics*, **5**, 169–85.

Carpenter, J., Matthews, P.H., and Schirm, J. (2010) Tournaments and office politics: evidence from a real effort experiment, *American Economic Review*, **100**, 504–17.

Chan, W., Courty, P., and Li, H. (2009) Suspense: dynamic incentives in sports contests, *The Economic Journal*, **119**, 24–46.

ChessBase. (2007) Blindfold Chess World Cup in Bilbao, http://en.chessbase.com/post/blindfold-che-world-cup-in-bilbao (accessed 14 August 2018).

Chevalier, J. and Ellison, G. (1997) Risk taking by mutual funds as a response to incentives, *Journal of Political Economy*, **105**, 1167–200.

Coupé, T., Smeets, V., and Warzynski, F. (2006) Incentives, sorting and productivity along the career: evidence from a sample of top economists, *Journal of Law, Economics, and Organization*, **22**, 137–67.

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Dilger, A. and Geyer, H. (2009) Are three points for a win really better than two? A comparison of German soccer league and cup games, *Journal of Sports Economics*, **10**, 305–18.

Dohmen, T. and Falk, A. (2011) Performance pay and multidimensional sorting: productivity, preferences, and gender, *American Economic Review*, **101**, 556–90.

Ehrenberg, R.G. and Bognanno, M.L. (1990) Do tournaments have incentive effects? *Journal of Political Economy*, **98**, 1307–24.

Fort, R. and Quirk, J. (1995) Cross-subsidization, incentives, and outcomes in professional team sports leagues, *Journal of Economic Literature*, **33**, 1265–99.

Freeman, R.B. and Gelber, A.M. (2010) Prize structure and information in tournaments, *American Economic Journal: Applied Economics*, **2**, 149–64.

Genakos, C. and Pagliero, M. (2012) Interim rank, risk taking, and performance in dynamic tournaments, *Journal of Political Economy*, **120**, 782–813.

Green, J.R. and Stokey, N.L. (1983) A comparison of tournaments and contracts, *Journal of Political Economy*, **91**, 349–64.

Guedes, J.C. and Machado, F.S. (2002) Changing rewards in contests: has the three-point rule brought more offense to soccer? *Empirical Economics*, **27**, 607–30.

Güth, W., Levínský, R., Pull, K., and Weisel, O. (2016) Tournaments and piece rates revisited: a theoretical and experimental study of output-dependent prize tournaments, *Review of Economic Design*, **20**, 69–88.

Haugen, K.K. (2008) Point score systems and competitive imbalance in professional soccer, *Journal of Sports Economics*, **9**, 191–210.

Knoeber, C.R. and Thurman, W.N. (1994) Testing the theory of tournaments: an empirical analysis of broiler production, *Journal of Labor Economics*, **12**, 155–79.

Lazear, E.P. (2000) Performance pay and productivity, *American Economic Review*, **90**, 1346–61.

Lazear, E.P. and Rosen, S. (1981) Rank order tournaments as optimum labor contracts, *Journal of Political Economy*, **89**, 841–64.

Lee, Y.H. and Parinduri, R.A. (2016) Does the three-point rule make soccer more exciting? Evidence from a regression discontinuity design, *Journal of Sports Economics*, **17**, 377–95.

Moschini, G. (2010) Incentives and outcomes in a strategic setting: the 3-points-for-a-win system in soccer, *Economic Inquiry*, **48**, 65–79.

Nalebuff, B.J. and Stiglitz, J.E. (1983) Prizes and incentives: towards a general theory of compensation and competition, *Bell Journal of Economics*, **14**, 21–43.

Nunn, J. (2005) The draw problem—a simple solution, *ChessBase*, http://en.chessbase.com/post/the-draw-problem-a-simple-solution (accessed 14 August 2018).

Oyer, P. (1998) Fiscal year ends and non-linear incentive contracts: the effect on business seasonality, *Quarterly Journal of Economics*, **113**, 149–85.

Palacios-Huerta, I. (2004) Structural changes during a century of the world's most popular sport, *Statistical Methods and Applications*, **13**, 241–58.

Sala, G., Burgoyne, A.P., Macnamara, B.N., Hambrick, D.Z., Campitelli, G., and Gobet, F. (2017) Checking the 'academic selection' argument. Chess players outperform non-chess players in cognitive skills related to intelligence: a meta-analysis, *Intelligence*, **61**, 130–9.

Sato, Y. (2016) Fund tournaments and asset bubbles, *Review of Finance*, **20**, 1383–426.

Shearer, B. (2004) Piece rates, fixed wages and incentives: evidence from a field experiment, *The Review of Economic Studies*, **71**, 513–34.

Shen, C.H. and Zhang, H. (2018) Tournament incentives and firm innovation, *Review of Finance*, **22**, 1515–48.

Shipov, S. (2011) Chess as a sport and the three-point system, *ChessBase*, http://en.chessbase.com/post/shipov-che-as-a-sport-and-the-three-point-system (accessed 14 August 2018).

Sonas, J. (2011) What exactly is the problem of draws in chess? *ChessBase*, http://en.chessbase.com/post/sonas-what-exactly-is-the-problem (accessed 14 August 2018).

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